Question			Answer	Marks	AO	Guidance
12			For AB, Newton's second law $8=5a$			
			acceleration is 1.6 ms^{-2}	**B1**	**3.1b**	Uses Newton's second law to calculate acceleration
			for AB $u=0,\ a=1.6,\ t=3$			
			$s=\frac{1}{2}at^2=\frac{1}{2}\times 1.6\times 3^2=7.2$ m	**M1**	**1.1a**	Uses *suvat* equation(s) and their a leading to a value for s
			velocity at B $v=at=1.6\times 3=4.8$ ms^{-1}	**M1**	**3.1b**	Uses *suvat* equation(s) and their a leading to a value for velocity at B
			for BC Newton's second law $8-28=5a$	**M1**	**3.1b**	Uses Newton's second law to calculate acceleration. Condone missing 8N force. Allow sign errors.
			acceleration is -4 ms^{-2}	**A1**	**1.1b**	soi
			for BC $u=4.8,\ v=0,\ a=-4$			
			$0^2=4.8^2-2\times 4s$	**M1**	**1.1a**	Uses *suvat* equation(s) and their a leading to a value for s.
			$s=2.88$ m	**A1**	**1.1b**	FT their negative a and their positive velocity at B
			distance AC is $7.2+2.88=10.08$ m	**A1**	**1.1b**	Allow 10 m
				[8]		